Benchmarking integration features in OpenTURNS

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Table of contents

Context and objectives

Context

- Given a function $g\colon \mathbb{R}^d \to \mathbb{R}^{d_Y}$ and a probability density f on $\mathbb{R}^d$, we are interested in computing integrals of the form:

$$\mathbb{E}\left[g(\boldsymbol{X})\right] = \int_{\mathbb{R}^d} g(\boldsymbol{x}) f(\boldsymbol{x})\, \mathrm{d}\boldsymbol{x}.$$

- OpenTURNS offers a wide range of integration algorithms, from classical quadrature to advanced high-dimensional methods.
- However, a systematic benchmark of these methods is lacking.

Objectives

- Provide an overview of the available integration methods in OpenTURNS.
- Establish benchmarks using a collection of test problems to evaluate the performance of these methods.
- Compare the efficiency of different approaches for polynomial chaos expansion and expectation estimation.

Based on [Baudin et al., 2025].

Gauss quadratures, sampling, and tensorization

Base classes

- `Sample`: A set of `n` points in dimension `d`.
- `WeightedExperiment`: An experiment based on a `Distribution` and an `int` size. Implements a `generateWithWeights()` method to create a quadrature rule: the points are in a `Sample` and the weights are in a `Point`.
- `ExperimentIntegration`: Creates a quadrature rule based on a weighted experiment. Provides the `integrate()` method.

Sampling methods (arbitrary dimension)

- `MonteCarloExperiment`: Random samples from a given distribution.
- `LowDiscrepancyExperiment`: Quasi-Monte Carlo sequences with low discrepancy.
- `LHSExperiment`: Latin Hypercube Sampling.
- `ImportanceSamplingExperiment`: Variance reduction using importance sampling.
- `FixedExperiment`: A fixed set of points and weights, useful for testing.

Gaussian quadrature and low discrepancy sequences

Gauss quadratures

- `GaussLegendre`: Tensorized algorithm (`Uniform` density).
- `GaussProductExperiment`: Integration with respect to any `Distribution`.
- `GaussKronrod`: Adaptive integration algorithm for vector-valued functions: $g\colon \mathbb{R} \to \mathbb{R}^d$. The choice of rule is in `GaussKronrodRule`.

Tensorization

- `TensorProductExperiment`: Full tensorization of arbitrary `WeightedExperiment`.
- `SmolyakExperiment`: Incomplete tensorization (Smolyak quadrature).

Low discrepancy sequences

Inheriting from `LowDiscrepancySequence`:

- `FaureSequence`
- `HaltonSequence`
- `ReverseHaltonSequence`
- `HaselgroveSequence`
- `SobolSequence`

Specific algorithms and the Cuba Library

Specific algorithms

- `FejerAlgorithm`: Fejér and Clenshaw-Curtis algorithm generating nested rules (useful in Smolyak quadrature).
- `IteratedQuadrature`: Multivariate integration with bounds defined by functions of lower dimensions.
- `FilonQuadrature`: For oscillating integrands.
- `SimplicialCubature`: Integration on a mesh (arbitrary input and output dimensions). Efficient in low dimensions when the domain is different from a hypercube, e.g., a banana-shaped domain[a].

[a] See [Daniel et al., 2025] in a Bayesian setting.

The Cuba library

The `CubaIntegration` class provides 4 multidimensional algorithms from the Cuba library:

- `Cuhre`: Deterministic and adaptive algorithm.
- `Divonne`: Stratified Monte Carlo sampling.
- `Suave`: Monte Carlo sampling combining importance sampling and subregion sampling.
- `Vegas`: Monte Carlo sampling combining importance sampling and variance reduction.

Optimization and space filling

LHS design optimization

Methods aimed at improving the distribution quality of an LHS design.

- `OptimalLHSExperiment`: Base class for optimal designs.
- `MonteCarloLHS`: Random search optimization.
- `SimulatedAnnealingLHS`: Simulated annealing optimization.
- `LHSResult`: Summary of optimization results.

Space filling criteria

Mathematical indicators used to evaluate the quality of a design.

- `SpaceFilling`: Base class for computation.
- `SpaceFillingC2`: Centered L^2 discrepancy.
- `SpaceFillingMinDist`: Minimal distance criterion.
- `SpaceFillingPhiP`: ϕ_p criterion.

Purpose: Can be used for kriging, but a LHS is a valid sampling method for integration.

Key considerations for integration methods

1. Input dimension and domain constraints

Most OpenTURNS methods compute integrals over hyper-rectangles, with notable exceptions:

- `SimplicialCubature` & `IteratedQuadrature`: Distinguish OpenTURNS by handling non-standard domains.
- `WeightedExperiment`: Integrates over the support of the underlying distribution (potentially infinite or exotic shapes).

2. Stopping criterion

Is the algorithm driven by:

- A **fixed budget** (number of evaluations)?
- A **target precision** (adaptive error control)?

Review of the use of test problems

Historical context:

- Using test problem collections to compare integration algorithms is a standard practice, e.g., [Davis and Rabinowitz, 2007, Genz, 1984].
- Past applications: evaluating Smolyak quadrature [Novak and Ritter, 1996] and the efficiency of scrambled QMC methods [Owen, 1998].
- Advantage: simple test functions are easy to implement and analyze [Faure and Lemieux, 2009, Owen, 2003].

Motivations for a new analysis:

- Literature limitation: studies often focus on narrow subsets of methods (e.g., comparing different low-discrepancy sequences [Faure and Lemieux, 2009] or Smolyak variants [Gerstner and Griebel, 1998]).
- **Lack of cross-comparisons**: few direct comparisons exist between QMC methods and Smolyak quadrature for high-dimensional integration.
- Objectives: compare these rarely juxtaposed approaches and establish reference benchmarks for the OpenTURNS library.

Fejér and Clenshaw-Curtis rules

Example 1

- Consider[a] the function $g(x) = |x - 0.3|$ for all $x \in [-1, 1]$.
- Assume that X has the uniform probability density between -1 and 1.
- We estimate the integral $\int_{-1}^{1} g(x) f(x) \, dx$.
- The coefficients of g in the Chebyshev decomposition decrease as $O(n^{-2})$.

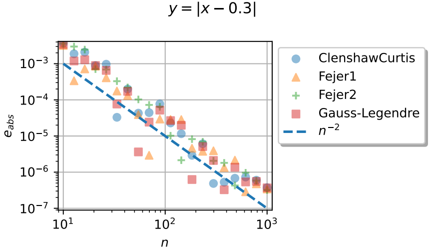

Figure 1: Absolute error against sample size for 4 quadrature rules: Clenshaw-Curtis, Fejér 1 and 2, and Gauss-Legendre.

Remark. The `FejerExperiment` class has a bug in OpenTURNS 1.26[b].

[a] See [Trefethen, 2012].
[b] See #3149: The weights in FejerExperiment are wrongly scaled.

Benchmark for expectation computation

Benchmark setup

- **Tested functions:** Ishigami and G-Sobol'.
- **Metric:** Absolute estimation error against sample size.
- **Methods:** MC, LHS, optimized LHS, Gauss, Smolyak, and Sobol' QMC.

Sobol' sequence adaptation

For $i \geq 0$, the choice of sample size n impacts sequence convergence:

- **Without adaptation:** $n = \lfloor n_0 \pi^i \rfloor$, where $n_0 \in \mathbb{N}$ is the initial size.
- **With adaptation:** $n = 2^i$, to optimize the convergence.

Expectation estimation: Ishigami function

Example 2

- The Ishigami function is[a]:

  $$g(\boldsymbol{x}) = \sin(x_1) + a\sin^2(x_2) + b x_3^4 \sin(x_1)$$

  where $a = 7$ and $b = 0.1$ for any $x_i \in [-\pi, \pi]$.
- $\boldsymbol{X} \sim \mathcal{U}([-\pi, \pi]^3)$.
- The sample size is a power of 2.

Conclusion: For the Ishigami function, the best methods are Gauss and Smolyak, if $n > 10^2$.

[a] See [Ishigami and Homma, 1990].

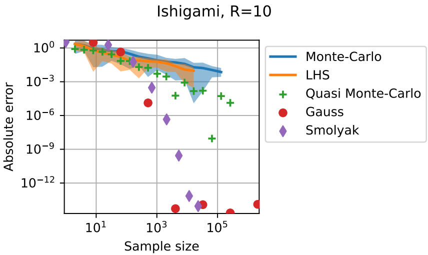

Figure 2: Absolute error of expectation estimation against sample size for the Ishigami function. Confidence bounds are based on 95% confidence intervals.

Expectation estimation: G-Sobol' function

Example 3

- The G-Sobol' function is[a]:

$$g(\boldsymbol{x}) = \prod_{i=1}^{d} g_i(x_i)$$

where d is the dimension and:

$$g_i(x_i) = \frac{|4x_i - 2| + a_i}{1 + a_i}$$

for any $x_i \in [0, 1]$.

- $\boldsymbol{X} \sim \mathcal{U}([0, 1]^d)$.
- Default parameters: $d = 3$ and $\boldsymbol{a} = (0, 9, 99)^\top$.
- The sample size is a power of 2.

[a]See [Saltelli and Sobol, 1995].

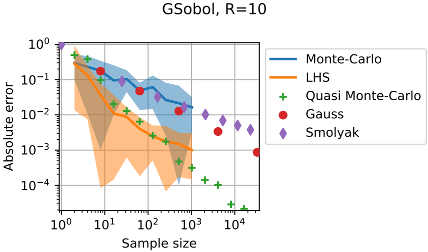

Figure 3: Absolute error of expectation estimation against sample size for the G-Sobol' function. Confidence bounds are based on 95% confidence intervals.

Conclusion: For the G-Sobol' function, the best method is quasi-Monte Carlo, if $n > 10^3$.

QMC Sobol' adaptation and sampling time

Expectation Estimation: QMC Sobol' Adaptation
Consider the G-Sobol' function with QMC and the Sobol' sequence.

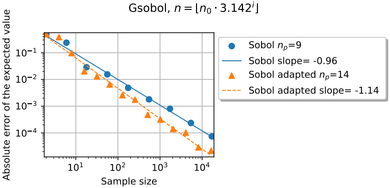

Figure 4: Absolute error for QMC Sobol' sampling, with and without adaptation, for the G-Sobol' function.

Conclusion: If the sample size is a power of 2, then the Sobol' sequence can be more accurate.

Measuring sampling time to generate the experimental design.

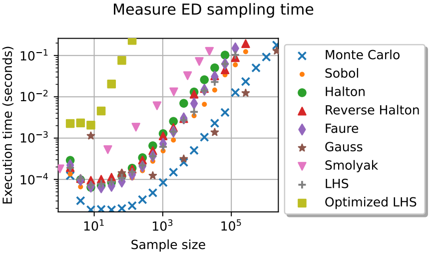

Figure 5: Elapsed time to evaluate the `generate()` method.

Conclusion: The optimized LHS takes more time than other methods. Monte Carlo sampling and Gaussian quadrature are faster.

Polynomial chaos: integration method

Coefficient estimation

- **Method:** Integration via quadrature[a].
- **Metric:** $\mathcal{L}_f^2$ error of the expansion.
- **Objective:** Quantify the impact of the sampling method on precision.

Application: Ishigami function

- Gauss quadrature achieves high precision with relatively few points.

[a] Alternative: Least squares.

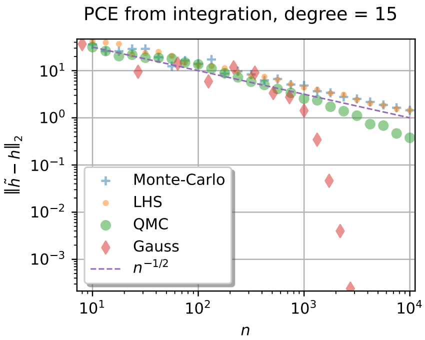

Figure 6: $\mathcal{L}_f^2$ error against sample size for various quadrature methods.

Polynomial chaos: quadrature sampling

Example 4 (Coefficient estimation by simple Monte Carlo for the mixed function)

- Consider the “mixed” model:

$$g(\boldsymbol{x}) = x_1 + \beta x_1 x_2$$

  for any $\boldsymbol{x} = (x_1, x_2)^\top \in \mathbb{R}^2$ where $\beta \in \mathbb{R}$ is a parameter.

- Assume independent variables $X_1 \sim \mathcal{N}(0, 0.1^2)$ and $X_2 \sim \mathcal{N}(1, 1^2)$, with $\beta = 2$.

Expand it as a polynomial chaos of total degree equal to 2.

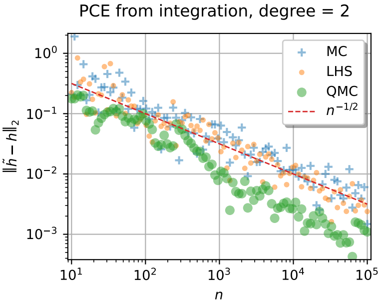

Figure 7: $\mathcal{L}_f^2$ error of the polynomial chaos expansion for the "*mixed*" function when the coefficients are estimated by quadrature with different sampling methods.

Conclusion: The QMC sampling method is faster here, if n is large enough.

Collection of test problems (`otintegr`)

The module provides 16 test problems[1].

Index	**Problem**	**Dim.**	**Index**	**Problem**	**Dim.**
0	CornerPeak	3	8	ProductPeak	3
1	C0	3	9	MorokoffCaflisch1	3
2	Discontinuous	3	10	MorokoffCaflisch2	3
3	Gaussian	3	11	Affine	3
4	GSobol	3	12	Flood1	1
5	Ishigami	3	13	Runge	1
6	AdditiveIshigami	3	14	Flood4	4
7	Oscillatory	3	15	Flood8	8

[1] See [Genz, 1984].

Borda score analysis

- A method's Borda score is its average score across all sample sizes.
- To compare absolute errors across common sample sizes, we use Nadaraya-Watson kernel regression in log-log space.
- The maximum elapsed time for a single point on the error curve is set to 0.5s.

Conclusion:

- The best method is quasi-Monte Carlo (Sobol' sequence), followed by Gauss quadrature.

⚠ Results depend on the chosen test problems and sample sizes.

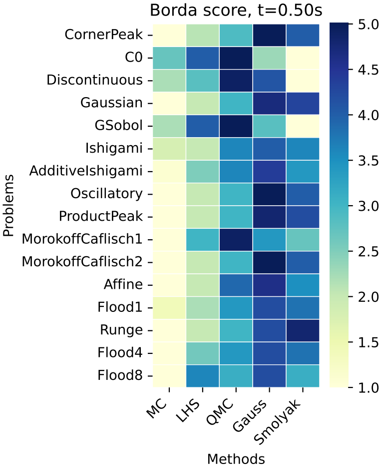

Figure 8: Average Borda score for each problem and method.

Limitations and conclusion

Limitations

- The mandatory `ot.Function` definition is unintuitive for users. Unclear error messages[a] and insufficient documentation[b] hinder usability.
- Quadrature methods completely fail if the computer code has **failures**.
- The first point of Sobol' sequence should be fixed[c].

[a] See #3108: Improve error message transparency in PythonFunction callbacks.
[b] See #1780: There is no example of OpenTURNSPythonFunction.
[c] See #1601: The first point of the Sobol' sequence is not zero.

Conclusions

- A comprehensive set of algorithms is available for integration, ranging from classical approaches (Monte Carlo, Gauss) to advanced high-dimensional methods (Smolyak, adaptive QMC, Cuba).
- A benchmark can be used to find bugs in the library[a].

Perspectives

- Incorporation of integration benchmark into `otbenchmark`[b].
- Implementation of D-optimal design of experiments[c].

[a] See #2653: The Faure sequence is wrong (closed).
[b] See [Fekhari et al., 2021].
[c] See [Baudin et al., 2025].

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